NYU Tandon School of Engineering

Algorithmic Portfolio Management

FRE7241 Homework 2

Volatility Timing and Hampel Strategies

Methods, results, and interpretation

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November 7, 2026

https://github.com/ph2606/fre7241-homework-02

Abstract

This assignment evaluates two rules using the supplied course data. Part I holds a position in SPY proportional to the inverse of its trailing exponentially weighted volatility, delays the position by one trading day, and deducts spread costs. A parallel parameter sweep compares nine decay factors. Part II applies a three-observation Hampel mean-reversion rule to individual stocks, averages their daily strategy PnLs, and examines the relationship between Sharpe ratios through 2016 and from 2017 onward. The top-100 portfolio is selected using the first period only. All curves show cumulative sums of daily PnLs. Across 645 stocks observed in both intervals, the Sharpe correlation is 0.0293, providing little evidence of persistence. The top-100 portfolio's Sharpe declines from 2.1925 in sample to 0.4027 afterward. The numerical results use frozen local data, whose dates and universe differ from the template's illustrative outputs.

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1 Data, conventions, and reproducibility

The completed submission is `Panagiotis_Housos_hw2.R`. The methods follow the course's volatility and Hampel examples [1, 2]. The accompanying repository supplies the data snapshot, numerical outputs, interactive plots, LaTeX source, and independent verification records:

https://github.com/ph2606/fre7241-homework-02

Part I uses the installed `rutils::etfenv$returns` SPY series after removing missing returns: 8,352 daily observations from February 1, 1993 through April 7, 2026. Part II uses the supplied `06_Data/R_Data/sp500_prices.RData`, restricted to dates in `"2006/"`. Its price matrix has 5,042 rows and 717 stock columns, ending January 16, 2026. The portable input snapshot is `data/hw2_inputs.rds`.

The professor's template illustrates a different data version: 8,298 SPY returns ending May 8, 2026, and approximately 1,000 stock strategies over 5,119 dates. These reference values are retained in the original template; the completed code reports the values calculated from the available inputs. Different dates, price revisions, and stock coverage can change every performance statistic and ranking. The reference figures establish the requested plot types.

The supplied SPY returns are daily log returns. Part II computes differences of log prices. Returns and PnLs are kept as decimal values, and `cumsum()` produces the plotted cumulative PnL. These sums are not compounded dollar wealth. The backtests assume zero risk-free return, 252 trading days per year, and one-observation position delays. The volatility estimator has the initialization exception documented below.

The course package version installed here exposes `HighFreq::run_var(timeser, lambdaf)`. The code passes the decay parameter positionally; the template's `lambda` spelling also matches the installed formal argument by R's partial-name matching. Package versions, input hashes, and session information are recorded under `reproducibility/`. This makes the analysis repeatable without the instructor's absolute data path.

2 Part I: inverse-volatility timing of SPY

2.1 I.1: positions, transaction costs, and risk adjustment

Let r_t denote SPY's daily log return. The course function `HighFreq::run_var()` calculates a trailing exponentially weighted mean and variance. For the initial calculation the decay factor is $\lambda = 0.5$. Following the installed implementation, its recursion after startup is

$$\widehat{\mu}_t = \lambda \widehat{\mu}_{t-1} + (1-\lambda) r_t, \tag{1}$$

$$\widehat{v}_t = \lambda^2 \widehat{v}_{t-1} + (1-\lambda^2)(r_t - \widehat{\mu}_t)^2. \tag{2}$$

The volatility estimate is the square root of the function's second output column:

$$\widehat{\sigma}_t = \sqrt{\widehat{v}_t}, \qquad u_t = \frac{1}{\widehat{\sigma}_t}, \qquad w_t = u_{t-1}, \quad w_1 = 0. \tag{3}$$

The one-day delay uses `rutils::lagit()`. The installed package initializes its first variance from the first *two* returns, $\widehat{v}_1 = (r_2 - r_1)^2$. Consequently, the second observation's held position uses its own return through this startup estimate. The assignment's estimator and reference position convention are

retained, so the second observation is not fully ex ante. After this initialization, the delayed recursive estimates use information available before the return to which the position is applied.

The inverse-volatility position increases after quiet periods and decreases after volatile periods. Its raw numerical size reflects reciprocal daily volatility; the final PnLs are rescaled below.

The assigned bid–ask spread is $b = 0.0001$. Trading at half the spread for each unit of position turnover gives

$$c_t = \frac{b}{2}|w_t - w_{t-1}|, \qquad q_t = w_t r_t - c_t, \tag{4}$$

with zero initial position and the course's zero-padded difference convention. A larger change in exposure incurs a larger cost. The net strategy is then matched to SPY's observed daily standard deviation:

$$a = \frac{s(r)}{s(q)}, \qquad p_t = a q_t. \tag{5}$$

The scalar a multiplies both trading gains and deducted costs. Since a uses the full sample, the volatility matching is a descriptive comparison. A trading implementation would estimate its scale from information already available at the time. A positive constant scale leaves the Sharpe and course-defined Sortino ratios unchanged.

The annualized Sharpe ratio is

$$\widehat{\mathrm{SR}}(x) = \sqrt{252}\,\frac{\overline{x}}{s(x)}. \tag{6}$$

The assignment's ratios use `rutils::calc_sharpe()`, whose second row uses the standard deviation of the subset of negative observations:

$$\widehat{\mathrm{Sortino}}_{\mathrm{course}}(x) = \sqrt{252}\,\frac{\overline{x}}{s(\{x_t : x_t < 0\})}. \tag{7}$$

This conditional downside standard deviation differs from downside semideviation computed across every date. The report uses the course definition consistently.

Table 1: SPY and the $\lambda = 0.5$ volatility-timing strategy. The strategy includes spread costs and is scaled to SPY's daily standard deviation. Annualization uses 252 trading days.

Series	Sharpe	Course Sortino	Annualized volatility	Mean daily PnL
SPY	0.5360	0.6689	0.1861	0.000396
Strategy	0.6548	0.7982	0.1861	0.000484

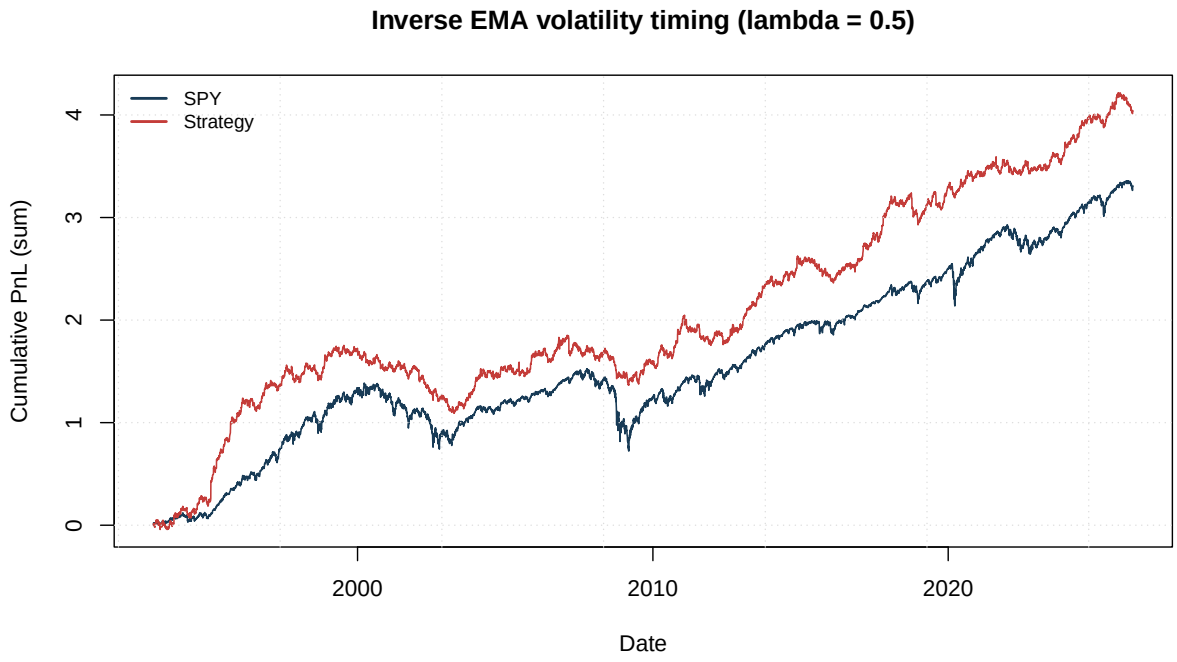


Figure 1: SPY and volatility-timing cumulative PnLs. The red strategy curve includes the spread costs in Equation 4; both series have the same full-sample daily standard deviation.

Table 1 compares average return relative to total and negative-return variation. At $\lambda = 0.5$, the net strategy Sharpe is 0.6548, compared with 0.5360 for SPY. The corresponding course Sortino ratios are 0.7982 and 0.6689. Figure 1 also shows the timing of gains and losses; its terminal height alone is not a risk-adjusted performance measure. The rule can remain exposed when volatility rises abruptly because yesterday's estimate determines today's position. The backtest does not charge financing costs or impose a leverage limit.

2.2 I.2: parallel sweep over decay factors

The sweep evaluates

$$\lambda \in \{0.1, 0.2, \ldots, 0.9\}. \tag{8}$$

For each decay factor, the code recomputes the variance, inverse-volatility positions, delayed positions, spread costs, and Sharpe ratio. A `parallel` worker cluster performs the nine independent calculations; the workers receive the same returns and cost assumptions. The cluster is stopped after the calculation. No random simulation is involved.

Table 2: Net Sharpe ratios for the nine decay factors. The maximum is selected with `max()` and its parameter with `which.max()`.

Decay factor λ	Net annualized Sharpe
0.1	0.5762
0.2	0.6733
0.3	0.6833
0.4	0.6712
0.5	0.6548
0.6	0.6323
0.7	0.6141
0.8	0.6116
0.9	0.6307

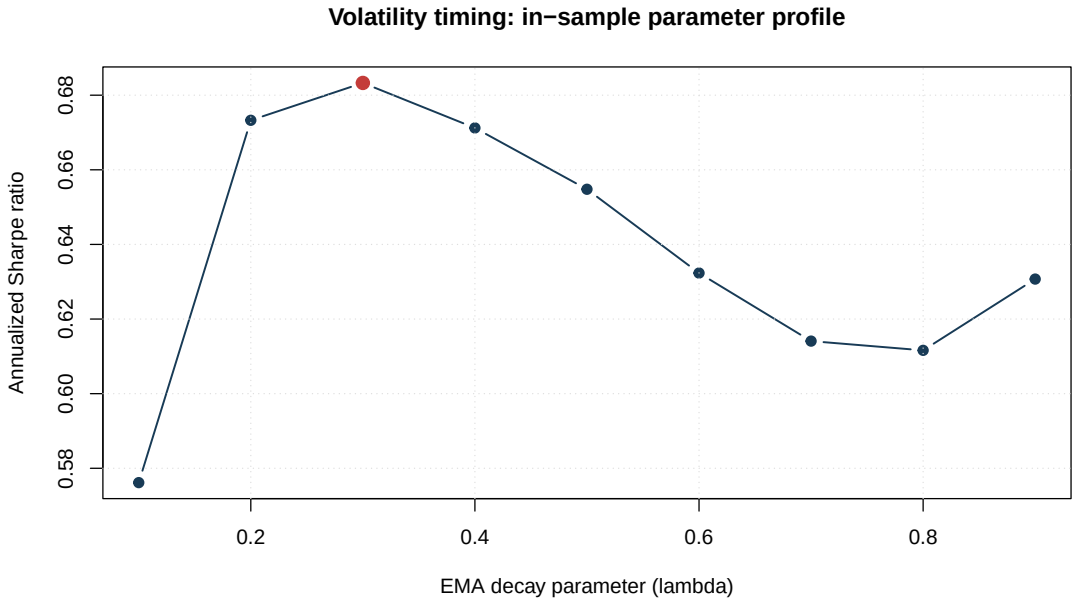


Figure 2: Net strategy Sharpe ratio versus the EMA decay factor. Every point uses the same complete historical SPY sample.

Smaller decay factors respond more quickly to recent returns and can produce greater turnover. Larger factors smooth the volatility estimate more heavily. Their Sharpe ratios reflect both this change in exposure and the costs of adjusting it. The highest ratio is 0.6833, attained at $\lambda = 0.3$. The maximum in Table 2 is an in-sample parameter choice among nine alternatives; it is not evidence of performance on an independent test sample.

3 Part II: Hampel strategies for individual stocks

3.1 II.1: stock-level signals and Sharpe rankings

An `lapply()` loop processes each stock separately. It removes missing prices for that stock and omits a series with fewer than $2 \times \texttt{lookb} = 6$ valid observations. Its PnL series retains the stock's actual dates, including warmup observations with zero positions. Removing missing prices means that consecutive valid observations may span more than one trading date for an interrupted series.

Let $\ell_{t,j} = \log P_{t,j}$ be stock j's log price. With a trailing window of $L = 3$ observations, define

$$m_{t,j} = \text{median}\{\ell_{t-L+1,j}, \ldots, \ell_{t,j}\}, \tag{9}$$

$$d_{t,j} = 1.4826\,\text{median}\{|\ell_{s,j} - m_{t,j}| : s = t - L + 1, \ldots, t\}, \tag{10}$$

$$z_{t,j} = \begin{cases} (\ell_{t,j} - m_{t,j})/d_{t,j}, & d_{t,j} > 0, \\ 0, & d_{t,j} = 0. \end{cases} \tag{11}$$

The factor 1.4826 is the normal-consistency multiplier used by the installed `HighFreq` nonparametric dispersion calculation. The code calls `roll_mean(..., method="nonparametric")` and `roll_var(..., method="nonparametric")` to obtain the trailing median and scaled MAD. The first L z-scores are set to zero.

The contrarian target position, with threshold one, is

$$h_{t,j} = \begin{cases} -1, & z_{t,j} > 1, \\ +1, & z_{t,j} < -1, \\ h_{t-1,j}, & -1 \le z_{t,j} \le 1, \end{cases} \qquad h_{1,j} = 0. \tag{12}$$

Thus a price sufficiently above its trailing median triggers a short position; a price sufficiently below it triggers a long position. A neutral signal preserves the previous position. The code fills forward the target positions, lags them by one observation, and calculates

$$r_{t,j} = \ell_{t,j} - \ell_{t-1,j}, \qquad \pi_{t,j} = h_{t-1,j} r_{t,j}, \tag{13}$$

using a zero first log-price difference. This retains zero warmup PnLs without a look-ahead signal. No transaction costs are deducted in Part II, as specified by the assignment's interpretation.

The resulting named list `pnls` stores a one-column `xts` PnL series for each of the 717 eligible stocks. Applying Equation 6 to each series and sorting in descending order produces the full-period Sharpe ranking.

Table 3: Ten largest full-period stock-level Hampel Sharpe ratios, calculated from the available stock universe. These ranks use each stock's own valid observation history.

Rank	Stock	Full-sample Sharpe
1	POM	1.4837
2	CMCSA	1.0898
3	MJN	0.9728
4	AMCR	0.9332
5	SHW	0.9293
6	GEHC	0.9263
7	SIAL	0.9029
8	LIN	0.8857
9	HCBK	0.8827
10	SOLV	0.8648

A high full-period Sharpe ratio here is a description of one stock's historical strategy returns. The lengths and dates of stock histories differ. These full-period rankings are not used to choose the out-of-sample top-100 portfolio in Section 3.4.

3.2 II.2: the mean strategy and the NXPI example

The stock PnLs are merged on the union of their dates. For each date t, let A_t contain stocks with an available PnL. The assignment's row mean is

$$\overline{\pi}_t = \frac{1}{|A_t|} \sum_{j \in A_t} \pi_{t,j}. \tag{14}$$

The code uses `rowMeans(..., na.rm=TRUE)` and preserves the date index in a one-column `xts` series. Zero strategy positions remain valid zero PnLs; unavailable stock observations are omitted from the average. Consequently, the number and identity of stocks in the daily mean can change over time. An all-missing row is marked missing rather than treated as a zero return. The available count ranges from 542 to 655 stocks across the observed dates, out of 717 stocks appearing somewhere in the input history.

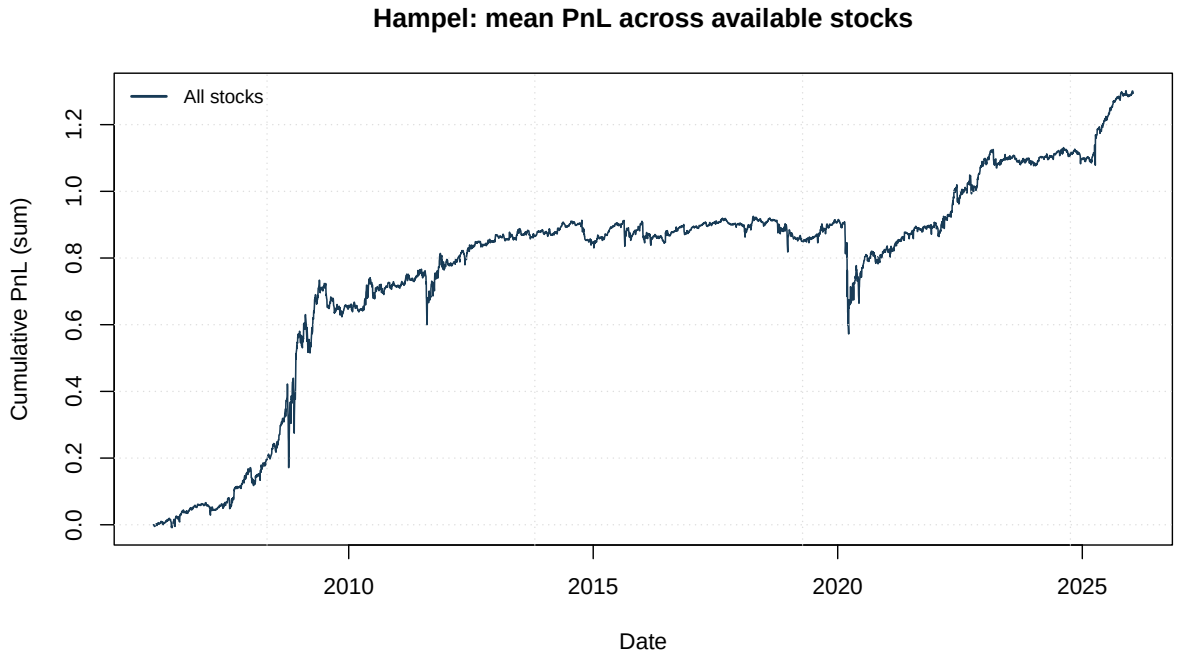


Figure 3: Cumulative mean daily Hampel PnL across all available stock strategies. The changing available-stock denominator follows Equation 14. No stock-strategy transaction costs are deducted.

The full-period Sharpe ratio of the all-stock mean PnL is 0.5415. Averaging individual PnLs reduces variation arising from imperfectly correlated stock strategies; this portfolio ratio is not the arithmetic average of their individual Sharpe ratios.

The NXPI plot aligns its buy-and-hold daily log returns with the dates of its Hampel strategy. Both cumulative curves begin with the same zero-return convention. Their Sharpe ratios use the same overlapping observations, so the comparison is not driven by unequal date samples.

Table 4: NXPI buy-and-hold and Hampel performance over the same available dates. Part II excludes transaction costs.

NXPI series	Sharpe	Course Sortino
NXPI	0.4509	0.6290
Hampel	0.6674	0.9246

NXPI's buy-and-hold Sharpe is 0.4509, compared with 0.6674 for the Hampel strategy. This individual example does not establish that the rule works equally well for other stocks.

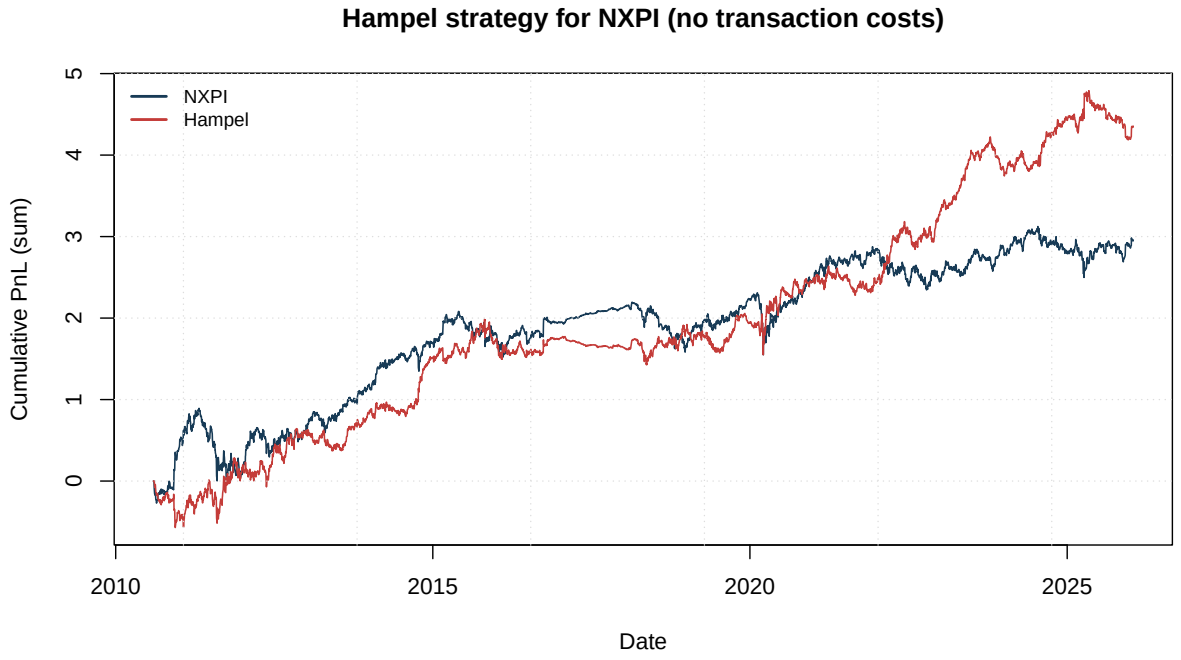


Figure 4: NXPI cumulative log returns and cumulative Hampel PnLs over NXPI's available history. The delayed strategy can hold long or short positions.

The course asks for interactive `dygraphs`. The R submission writes the requested interactive charts as HTML files alongside the corresponding static report figures. The static plots allow the report to remain self-contained when submitted as a PDF.

3.3 II.3: in-sample and out-of-sample Sharpe persistence

The sample split is fixed at the end of 2016. The in-sample interval includes dates through December 31, 2016; the out-of-sample interval begins January 1, 2017. Signals are generated sequentially on each stock's complete history, so the first 2017 position can use the final observed information from 2016. The split does not reset the trading position or recalculate past signals.

For each stock, the code calculates Sharpe ratios separately in the two intervals and retains stocks with finite ratios in both. In particular, stocks that began trading after 2016 cannot enter the comparison. Matching by ticker places the same stocks in the same order in `sharpis` and `sharpos`. The finite-ratio condition does not impose a long minimum history within each interval. A stock observed for only a short part of a subperiod can therefore have an unstable ratio and appear as an extreme point. The scatterplot fits

$$\widehat{\mathrm{SR}}_{j,\mathrm{OS}} = \widehat{\alpha} + \widehat{\beta}\widehat{\mathrm{SR}}_{j,\mathrm{IS}} + e_j. \tag{15}$$

Table 5: Cross-sectional relationship between stock Hampel Sharpe ratios through 2016 and from 2017 onward. Only stocks with finite ratios in both periods enter the regression.

Statistic	Value
Matched stock pairs	645
Correlation	0.0293
Intercept	0.1053
Slope	0.0329
Conventional slope p-value	0.4572
R^2	0.000860

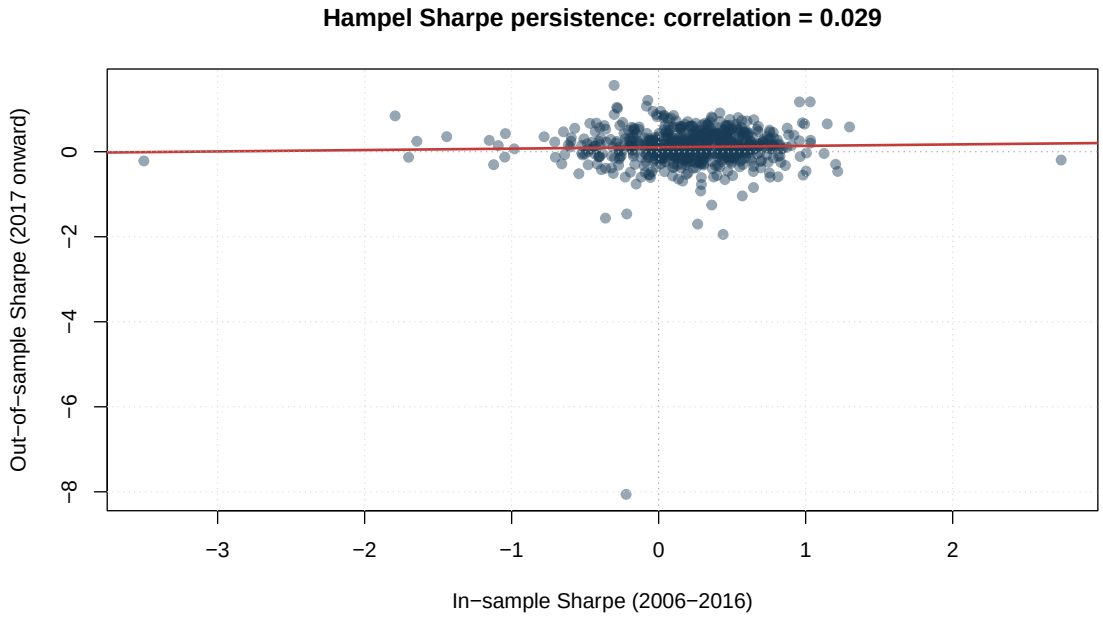


Figure 5: In-sample versus out-of-sample Hampel Sharpe ratios, with an ordinary least-squares regression line. Each point represents the same stock in the two periods.

The fitted slope is 0.0329, the correlation is 0.0293, and $R^2 = 0.000860$. The conventional slope p-value is 0.4572. The fitted relationship explains very little of the out-of-sample variation, and the available data provide little evidence of positive persistence. This differs from the template's comment and reference figure, which use a different stock dataset. A small positive fitted slope alone does not justify a claim that large in-sample Sharpe ratios reliably predict large later ratios.

Stocks are related through market and sector exposures, so a conventional regression p-value should not be interpreted as if all stock observations were independent experiments. The selected universe and unequal history lengths are further limits on generalizing this comparison.

3.4 II.4: the top-100 in-sample selection

The selected tickers are the first 100 names after sorting *in-sample* Sharpe ratios in descending order. Their full-history PnL series are then averaged using the same available-stock convention as Equation 14. No observation from 2017 onward is used to rank the stocks, and later data availability is not used to

filter the in-sample selection. The plotted full history contains both the period used to choose them and the subsequent evaluation period. The ticker selection is fixed, while its daily average includes only selected stocks with data on that date; a selected stock without later observations does not contribute to the later mean. In this snapshot, four selected tickers—SIAL, HCBK, SWY, and FDO—have no out-of-sample prices, leaving 96 selected stocks that appear during the later period.

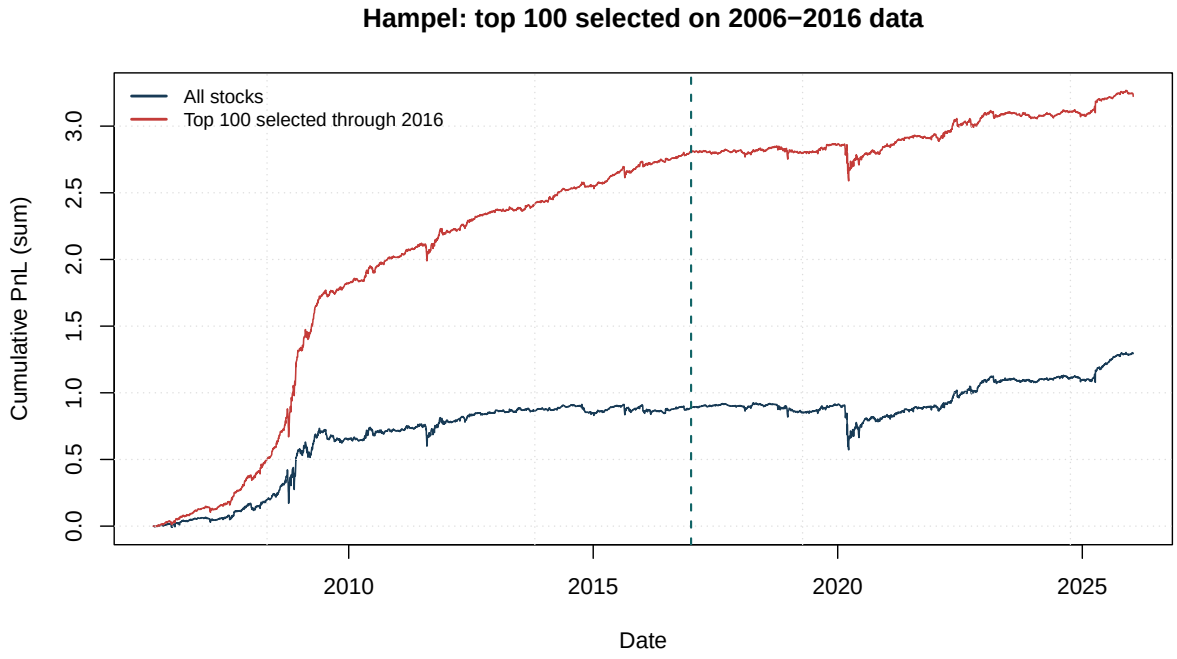


Figure 6: Cumulative mean Hampel PnL for the 100 stocks with the highest in-sample Sharpe ratios. The vertical split marks the start of 2017. The earlier segment is selected in sample; the later segment evaluates the available members of that fixed ticker selection.

Table 6: Mean-portfolio performance for all stocks and for the top 100 selected using only data through 2016. The separate out-of-sample rows evaluate dates from 2017 onward. Cumulative PnL is a sum of daily strategy PnLs.

Portfolio	Period	N	Sharpe	Cumulative PnL	Annualized volatility
All stocks	Full	5042	0.5415	1.2940	0.1194
Top 100	Full	5042	1.3852	3.2231	0.1163
All stocks	IS	2769	0.6653	0.8834	0.1208
Top 100	IS	2769	2.1925	2.8021	0.1163
All stocks	OOS	2273	0.3866	0.4106	0.1177
Top 100	OOS	2273	0.4027	0.4209	0.1159

The top-100 mean portfolio's annualized Sharpe falls from 2.1925 in sample to 0.4027 out of sample. Selection favors stocks with unusually good in-sample realizations, so the early top-100 curve is optimistically selected by construction. The later rows in Table 6 show that the striking early performance

does not carry forward at a comparable risk-adjusted level. These results still use the course's supplied stock universe and omit spread, slippage, stock-borrow, and financing costs. Short positions and frequent position reversals can make these omissions material. Historical membership and delisted-stock coverage are not established by the input file, so the analysis cannot exclude survivorship or universe-selection bias.

4 Numerical checks and deliverables

The verification record reports 63 passing checks. It checks calculations using independent reconstructions of the position delays, turnover costs, volatility scaling, and performance ratios. It also checks the parallel parameter sweep against a serial calculation and verifies the Hampel positions, date alignment, sample split, and restriction of top-100 ranking inputs to the first period.

The completed R file implements all six tasks, with a labeled section for each question. The original `FRE7241_hw2.R` is retained separately. The repository stores the inputs and the derived numerical tables so that the figures and report can be regenerated. The report's tables are exported from computed results instead of transcribed from the template. The README gives the package setup, R execution, and LaTeX build commands. The principal submission artifacts are:

- `Panagiotis_Housos_hw2.R`, the completed R assignment;
- `report/Panagiotis_Housos_HW2_Report.pdf`, this report;
- the public repository at https://github.com/ph2606/fre7241-homework-02.

The parameter sweep is in sample, Part I's volatility scale uses the full sample, and its two-return startup has the initialization exception described above. Part II's time split gives a later comparison period for stock selection, but costs and historical universe membership remain limitations. These are historical simulations rather than validated investment recommendations.

References

[1] J. Pawlowski. *Univariate Investment Strategies*, FRE7241 course slides, January 20, 2026, especially the trailing volatility example on slide 27 and the Hampel filter examples on slides 88–90.

[2] J. Pawlowski. `app_hampel_simple.R`, supplied course Shiny example under `05_Code_Examples/Shiny_Apps/`. The submitted assignment follows the lecture's log-price formulation.

[3] J. Pawlowski. `FRE7241_hw2.R` and six supplied reference figures. Homework 2, due November 10, 2026, at 6 p.m.

[4] Installed `rutils` and `HighFreq` implementations. Exact package versions and source information accompany the submission under `reproducibility/`.

A Complete submitted R code

The submission labels and completes all six questions. The original assignment and its illustrative output comments are supplied separately as `FRE7241_hw2.R`. Actual computed outputs use the frozen inputs documented in Section 1.

```
##################################
### FRE7241 Homework #2
### Panagiotis Housos | November 7, 2026
### Due: 6PM Tuesday November 10, 2026
##################################
# Run: Rscript --vanilla Panagiotis_Housos_hw2.R
# This file completes all six questions in FRE7241_hw2.R.
# Figures, interactive dygraphs, and numerical tables go to hw2_results/.
# The bundled input snapshot takes priority, so a clone reproduces the report.
# If only this R file is copied into the course's Homework_02 folder, it also
# runs using installed rutils data and ../../06_Data/R_Data/sp500_prices.RData.
# Different data/package snapshots can produce different numerical results.

suppressPackageStartupMessages(library(rutils))
stopifnot(requireNamespace("HighFreq", quietly = TRUE),
          requireNamespace("htmlwidgets", quietly = TRUE))

# Locate the submission without requiring the professor's absolute paths.
hw2_files <- c(sub("^--file=", "", grep("^--file=", commandArgs(), value = TRUE)),
               unlist(lapply(sys.frames(), function(x) x$ofile)))
hw2_files <- hw2_files[basename(hw2_files) == "Panagiotis_Housos_hw2.R"]
hw2_root <- if (length(hw2_files)) {
  dirname(normalizePath(tail(hw2_files, 1), mustWork = TRUE))
} else if (file.exists("Panagiotis_Housos_hw2.R")) {
  normalizePath(".")
} else {
  normalizePath(file.path("Homeworks", "Homework_02"), mustWork = TRUE)
}
results_dir <- file.path(hw2_root, "hw2_results")
for (subdir in c("", "tables", "interactive")) {
  dir.create(file.path(results_dir, subdir), recursive = TRUE, showWarnings = FALSE)
}
snapshot_file <- file.path(hw2_root, "data", "hw2_inputs.rds")
if (file.exists(snapshot_file)) {
  hw2_inputs <- readRDS(snapshot_file)
} else {
  stock_candidates <- c(
    file.path(hw2_root, "..", "..", "06_Data", "R_Data", "sp500_prices.RData"),
    file.path(hw2_root, "data", "sp500_prices.RData"),
    file.path(hw2_root, "sp500_prices.RData"),
    "/Users/jerzy/Develop/lecture_slides/data/sp500_prices.RData")
  stock_candidates <- stock_candidates[file.exists(stock_candidates)]
  if (!length(stock_candidates)) {
    stop("Missing stock data: use the repository data/ folder or course folder.")
  }
  stock_file <- stock_candidates[1]
  stock_env <- new.env()
  load(stock_file, envir = stock_env)
  hw2_inputs <- list(
    spy_returns = na.omit(get("SPY", rutils::etfenv$returns)),
    stock_prices = stock_env$pricestock["2006/"]
  )
}

# The course's Sortino uses SD of negative returns, not semideviation.
sharpe_ratio <- function(x) {
  x <- as.numeric(x)
  x <- x[is.finite(x)]
  if (length(x) < 2L || sd(x) == 0) return(NA_real_)
  sqrt(252) * mean(x) / sd(x)
}
course_ratios <- function(x) rutils::calc_sharpe(x, riskf = 0, nperiods = 252)
```

```
write_table <- function(x, filename, row_names = FALSE) {
  write.csv(x, file.path(results_dir, "tables", filename), row.names = row_names)
}

# Save each requested dygraph, including local JS/CSS dependencies.
save_dygraph <- function(x, filename, title) {
  widget <- dygraphs::dygraph(cumsum(x), main = title,
                              xlab = "Date", ylab = "Cumulative PnL (sum)")
  widget <- dygraphs::dyOptions(widget, colors = c("#183B56", "#C43C39"),
                                strokeWidth = 1.25, drawGrid = TRUE)
  widget <- dygraphs::dyRangeSelector(widget)
  htmlwidgets::saveWidget(widget,
    file.path(results_dir, "interactive", paste0(filename, ".html")),
    selfcontained = FALSE, libdir = "lib")
  widget
}

# Save report-quality PDF and PNG versions alongside interactive output.
save_figure <- function(filename, draw) {
  pdf(file.path(results_dir, paste0(filename, ".pdf")), width = 9, height = 5.3)
  tryCatch(draw(), finally = dev.off())
  png(file.path(results_dir, paste0(filename, ".png")),
      width = 1800, height = 1060, res = 200)
  tryCatch(draw(), finally = dev.off())
}
plot_pnl <- function(x, title, boundary = FALSE) {
  y <- coredata(cumsum(x))
  matplot(as.Date(index(x)), y, type = "l", lty = 1, lwd = 1.1,
          col = c("#183B56", "#C43C39"), xlab = "Date",
          ylab = "Cumulative PnL (sum)", main = title)
  grid(col = "#DDDDDD")
  if (boundary) abline(v = as.numeric(as.Date("2017-01-01")),
                       lty = 2, col = "#116466", lwd = 1.5)
  legend("topleft", legend = colnames(x), lty = 1, lwd = 2,
         col = c("#183B56", "#C43C39"), bty = "n", cex = 0.85)
}

############### Part I: inverse EMA volatility timing

bidask <- 0.0001
symboln <- "SPY"
retp <- hw2_inputs$spy_returns
# The installed HighFreq argument is 'lambdaf'; positional use also works
# with versions that call it 'lambda'. Column 2 is the EMA variance.
volma <- sqrt(HighFreq::run_var(as.matrix(retp), 0.5)[, 2])
stopifnot(all(is.finite(volma)), all(volma > 0))

# I.1 (10pts): inverse volatility, lagged positions, costs, and ratios.
posv <- rutils::lagit(1 / volma)
turnover <- abs(rutils::diffit(posv))
costs <- 0.5 * bidask * turnover
pnls_raw <- posv * retp - costs
vol_scale <- as.numeric(sd(retp) / sd(pnls_raw))
pnls <- pnls_raw * vol_scale
vol_pnls <- pnls
vol_metrics <- course_ratios(cbind(SPY = retp, Strategy = pnls))
colnames(vol_metrics) <- c("SPY", "Strategy")
cat("\nPart I.1: inverse volatility positions\n")
print(NROW(posv)); print(head(posv)); print(tail(posv))
cat("Net, volatility-matched PnLs\n")
print(head(pnls)); print(tail(pnls)); print(vol_metrics)
vol_plot_data <- cbind(SPY = retp, Strategy = pnls)
colnames(vol_plot_data) <- c("SPY", "Strategy")
dy_voltiming <- save_dygraph(vol_plot_data, "strat_voltiming_ema",
  sprintf("Volatility timing: SPY Sharpe %.3f / strategy %.3f",
          vol_metrics[1, 1], vol_metrics[1, 2]))
save_figure("strat_voltiming_ema", function() {
  plot_pnl(vol_plot_data, "Inverse EMA volatility timing (lambda = 0.5)")
})

```

```r
# I.2 (10pts): parallel lambda profile. PSOCK works on Windows and Unix.
library(parallel)
lambdav <- seq(0.1, 0.9, by = 0.1)
timing_sharpe <- function(lambda, retp, bidask) {
  vol <- sqrt(HighFreq::run_var(as.matrix(retp), lambda)[, 2])
  pos <- rutils::lagit(1 / vol)
  pnl <- pos * retp - 0.5 * bidask * abs(rutils::diffit(pos))
  # Positive constant volatility scaling leaves Sharpe unchanged.
  sqrt(252) * as.numeric(mean(pnl) / sd(pnl))
}
profile_parallel <- function(lambdav, retp, bidask) {
  workers <- min(2L, parallel::detectCores(logical = FALSE))
  if (is.na(workers)) workers <- 2L
  # Workers run without project startup profiles.
  cluster <- parallel::makePSOCKcluster(workers,
    rscript_args = "--vanilla", methods = FALSE)
  on.exit(parallel::stopCluster(cluster), add = TRUE)
  unlist(parallel::parLapply(cluster, lambdav, timing_sharpe,
                             retp = retp, bidask = bidask), use.names = FALSE)
}
sharpev <- profile_parallel(lambdav, retp, bidask)
lambda_profile <- data.frame(lambda = lambdav, Sharpe = sharpev)
best_sharpe <- max(sharpev)
best_lambda <- lambdav[which.max(sharpev)]
cat("\nPart I.2: parallel lambda profile\n")
print(round(sharpev, 4)); print(best_sharpe); print(best_lambda)
save_figure("strat_voltiming_profile_ema", function() {
  plot(lambdav, sharpev, type = "b", pch = 19, col = "#183B56", lwd = 1.5,
       xlab = "EMA decay parameter (lambda)", ylab = "Annualized Sharpe ratio",
       main = "Volatility timing: in-sample parameter profile")
  grid(col = "#DDDDDD")
  points(best_lambda, best_sharpe, pch = 19, col = "#C43C39", cex = 1.4)
})

############### Part II: Hampel strategy for single stocks

pricestock <- hw2_inputs$stock_prices["2006/"]
lookb <- 3L

# Course strategies_univariate.pdf pp. 88-90: use log prices, trailing
# median and scaled MAD; trade beyond +/-1, carry positions, then lag.
simulate_hampel <- function(prices, lookb = 3L, details = FALSE) {
  prices <- na.omit(prices)
  if (NROW(prices) < 2L * lookb) return(NULL)
  stopifnot(all(is.finite(prices)), all(prices > 0))
  pricev <- log(prices)
  nrows <- NROW(pricev)
  medianv <- HighFreq::roll_mean(as.matrix(pricev),
                                 lookb = lookb, method = "nonparametric")
  madv <- HighFreq::roll_var(as.matrix(pricev),
                             lookb = lookb, method = "nonparametric")
  zscores <- ifelse(madv > 0, (as.matrix(pricev) - medianv) / madv, 0)
  zscores[seq_len(lookb), ] <- 0
  signal <- rep(NA_integer_, nrows)
  signal[1] <- 0L
  signal[zscores > 1] <- -1L
  signal[zscores < -1] <- 1L
  signal <- as.numeric(zoo::na.locf(signal))
  pos <- rutils::lagit(signal)
  returns <- rutils::diffit(pricev)
  pnl <- returns * pos
  if (details) return(list(prices = pricev, median = medianv, mad = madv,
    zscores = zscores, signal = signal, positions = pos,
    returns = returns, pnl = pnl))
  pnl
}

# II.1 (10pts): one lapply() per column, omitting short histories.
pnls <- lapply(seq_len(NCOL(pricestock)), function(j) {
  simulate_hampel(pricestock[, j], lookb = lookb)
```

```
})
names(pnls) <- colnames(pricestock)
pnls <- Filter(Negate(is.null), pnls)
sharpev <- sort(vapply(pnls, sharpe_ratio, numeric(1)), decreasing = TRUE)
hampel_sharpe <- sharpev
cat("\nPart II.1: individual-stock PnLs and Sharpe ranking\n")
print(class(pnls)); print(dim(pnls[[1]]))
print(NROW(sharpev)); print(head(sharpev, 10))

# II.2 (10pts): date-aligned, equal-weight mean among available histories.
# merge() aligns dates; column binding by row number would mix listing dates.
pnl_all <- do.call(merge, c(unname(pnls), list(all = TRUE)))
colnames(pnl_all) <- names(pnls)
mean_available <- function(x) {
  values <- rowMeans(x, na.rm = TRUE)
  values[!is.finite(values)] <- NA_real_
  xts::xts(values, order.by = index(x))
}
pnlm <- mean_available(pnl_all)
colnames(pnlm) <- "All stocks"
available_counts <- rowSums(is.finite(pnl_all))
cat("\nPart II.2: mean PnLs\n")
print(class(pnlm)); print(dim(pnlm))
dy_hampel_mean <- save_dygraph(pnlm, "strat_hampel_mean",
  sprintf("Mean Hampel PnL: annualized Sharpe %.3f", sharpe_ratio(pnlm)))
save_figure("strat_hampel_mean", function() {
  plot_pnl(pnlm, "Hampel: mean PnL across available stocks")
})

symboln <- "NXPI"
nxpi_details <- simulate_hampel(pricestock[, symboln], lookb, details = TRUE)
nxpi_plot_data <- cbind(nxpi_details$returns, nxpi_details$pnl)
colnames(nxpi_plot_data) <- c("NXPI", "Hampel")
nxpi_metrics <- course_ratios(nxpi_plot_data)
dy_hampel_single <- save_dygraph(nxpi_plot_data, "strat_hampel_single",
  sprintf("NXPI: buy/hold Sharpe %.3f / Hampel %.3f",
          nxpi_metrics[1, 1], nxpi_metrics[1, 2]))
save_figure("strat_hampel_single", function() {
  plot_pnl(nxpi_plot_data, "Hampel strategy for NXPI (no transaction costs)")
})

# II.3 (10pts): in sample through 2016; out of sample from 2017.
# Run signals continuously, so the first 2017 holding uses 2016 information.
sharpis_all <- vapply(pnls, function(x) sharpe_ratio(x["/2016"]), numeric(1))
sharpos_all <- vapply(pnls, function(x) sharpe_ratio(x["2017/"]), numeric(1))
matched <- is.finite(sharpis_all) & is.finite(sharpos_all)
sharpis <- sharpis_all[matched]
sharpos <- sharpos_all[names(sharpis)]
stopifnot(identical(names(sharpis), names(sharpos)))
persistence_data <- data.frame(symbol = names(sharpis),
                               InSample = as.numeric(sharpis),
                               OutOfSample = as.numeric(sharpos))
persistence_fit <- lm(OutOfSample ~ InSample, data = persistence_data)
persistence_summary <- summary(persistence_fit)
persistence_correlation <- cor(sharpis, sharpos)
cat("\nPart II.3: matched in/out-of-sample Sharpe ratios\n")
print(NROW(sharpis)); print(head(sharpis, 10))
print(NROW(sharpos)); print(head(sharpos, 10)); print(persistence_summary)
save_figure("strat_hampel_scatter", function() {
  plot(sharpis, sharpos, pch = 19, col = adjustcolor("#183B56", alpha.f = 0.45),
       xlab = "In-sample Sharpe (2006-2016)",
       ylab = "Out-of-sample Sharpe (2017 onward)",
       main = sprintf("Hampel Sharpe persistence: correlation = %.3f",
                      persistence_correlation))
  grid(col = "#DDDDDD")
  abline(persistence_fit, col = "#C43C39", lwd = 2)
  abline(h = 0, v = 0, col = "#AAAAAA", lty = 3)
})
# A positive fitted slope describes cross-sectional historical persistence;
# common market exposures and selection prevent a causal interpretation.
```

```
273
274 # II.4 (10pts): choose the top 100 using ONLY in-sample Sharpe ratios.
275 ranked_is <- sort(sharpis_all[is.finite(sharpis_all)], decreasing = TRUE)
276 top_symbols <- head(names(ranked_is), 100L)
277 stopifnot(length(top_symbols) == 100L)
278 pnltop <- mean_available(pnl_all[, top_symbols])
279 colnames(pnltop) <- "Top 100 (selected through 2016)"
280 top_plot_data <- cbind(pnlm, pnltop)
281 colnames(top_plot_data) <- c("All stocks", "Top 100 selected through 2016")
282 dy_hampel_top <- save_dygraph(top_plot_data, "strat_hampel_top",
283   "Top 100 by in-sample Sharpe: selection ends December 2016")
284 dy_hampel_top <- dygraphs::dyEvent(dy_hampel_top, "2017-01-01",
285                                    "Out of sample", strokePattern = "dashed")
286 htmlwidgets::saveWidget(dy_hampel_top,
287   file.path(results_dir, "interactive", "strat_hampel_top.html"),
288   selfcontained = FALSE, libdir = "lib")
289 save_figure("strat_hampel_top", function() {
290   plot_pnl(top_plot_data, "Hampel: top 100 selected on 2006-2016 data", TRUE)
291 })
292
293 # Historical selected-stock gains before 2017 are in-sample. Evaluate the
294 # fixed selection after 2016 separately. Costs and shorting fees are excluded
295 # in Part II, as in the assignment; the supplied universe may be survivorship
296 # biased. Cumulative plots are sums of daily PnL, not compounded wealth.
297 portfolio_metrics <- do.call(rbind, lapply(c("Full", "IS", "OOS"), function(period) {
298   dates <- switch(period, Full = "/", IS = "/2016", OOS = "2017/")
299   do.call(rbind, lapply(c("All stocks", "Top 100"), function(portfolio) {
300     x <- if (portfolio == "All stocks") pnlm[dates] else pnltop[dates]
301     data.frame(Portfolio = portfolio, Period = period,
302       N = NROW(x), Sharpe = sharpe_ratio(x), CumulativePnL = sum(x),
303       AnnualizedVolatility = sqrt(252) * sd(x))
304   }))
305 }))
306 print(portfolio_metrics)
307
308 # Export the report's numerical inputs and reproducible result objects.
309 write_table(data.frame(Statistic = rownames(vol_metrics), vol_metrics,
310                        check.names = FALSE), "vol_ratios.csv")
311 write_table(lambda_profile, "lambda_profile.csv")
312 write_table(data.frame(Symbol = names(sharpev), Sharpe = as.numeric(sharpev)),
313             "hampel_sharpe_ranking.csv")
314 write_table(persistence_data, "hampel_is_oos.csv")
315 write_table(data.frame(Symbol = top_symbols, InSampleSharpe = ranked_is[top_symbols]),
316             "top100_selection.csv")
317 write_table(portfolio_metrics, "portfolio_metrics.csv")
318 write_table(data.frame(Date = as.character(index(vol_plot_data)),
319                        coredata(vol_plot_data)), "vol_daily_pnls.csv")
320 write_table(data.frame(Date = as.character(index(top_plot_data)),
321                        coredata(top_plot_data), AvailableStocks = available_counts),
322             "hampel_portfolio_daily_pnls.csv")
323 hw2_results <- list(
324   vol = list(returns = retp, volma = volma, positions = posv,
325     costs = costs, raw_pnls = pnls_raw, scale = vol_scale,
326     pnls = vol_pnls, metrics = vol_metrics, profile = lambda_profile,
327     best_lambda = best_lambda, best_sharpe = best_sharpe),
328   hampel = list(prices_dim = dim(pricestock), lookb = lookb,
329     first_pnl_dim = dim(pnls[[1]]), sharpe = sharpev,
330     mean_pnl = pnlm, available = available_counts,
331     nxpi = nxpi_details, nxpi_metrics = nxpi_metrics,
332     sharpis = sharpis, sharpos = sharpos,
333     regression = persistence_fit, correlation = persistence_correlation,
334     top_symbols = top_symbols, ranked_is = ranked_is,
335     top_pnl = pnltop, portfolio_metrics = portfolio_metrics)
336 )
337 saveRDS(hw2_results, file.path(results_dir, "results.rds"), compress = "xz")
338 cat("\nCompleted all six questions. Results saved in:\n", results_dir, "\n")
```

B Independent verification record

```
Homework 2 independent verification
Inputs: data/hw2_inputs.rds
Results: hw2_results/results.rds

PASS 01: Required numerical/time-series packages are available
PASS 02: Frozen input and saved result files exist
PASS 03: SPY input is finite and has sufficient observations
PASS 04: Independent EMA mean recurrence matches a known probe
PASS 05: EMA variance uses squared decay and first squared difference
PASS 06: SPY EMA means match independent recurrence
PASS 07: SPY EMA variances match independent recurrence
PASS 08: Saved EMA volatilities match independent recurrence
PASS 09: Inverse volatility positions are lagged one day
PASS 10: Costs equal half-spread times absolute position change
PASS 11: Net PnLs subtract turnover costs before normalization
PASS 12: Saved full-sample volatility scale is correct
PASS 13: Saved normalized timing PnLs match independent arithmetic
PASS 14: Timing volatility equals SPY volatility
PASS 15: First timing holding and PnL are flat
PASS 16: Timing PnL dates align with SPY returns
PASS 17: Sharpe and course Sortino match independent formulas
PASS 18: Saved PSOCK lambda profile matches independent sequential recurrence
PASS 19: Best Sharpe is profile maximum
PASS 20: Best lambda is associated with the maximum Sharpe
PASS 21: EMA prefix invariant at observation 100
PASS 22: Future SPY changes preserve EMA history through 100
PASS 23: EMA prefix invariant at observation 4176
PASS 24: Future SPY changes preserve EMA history through 4176
PASS 25: EMA prefix invariant at observation 8332
PASS 26: Future SPY changes preserve EMA history through 8332
PASS 27: Saved stock dimensions and lookback match frozen inputs
PASS 28: NXPI is present in the frozen universe
PASS 29: NXPI uses logarithmic prices
PASS 30: NXPI trailing medians match base median loop
PASS 31: NXPI scaled trailing MAD matches base mad after startup
PASS 32: NXPI independent base loop: zscores
PASS 33: NXPI independent base loop: signal
PASS 34: NXPI independent base loop: positions
PASS 35: NXPI independent base loop: returns
PASS 36: NXPI independent base loop: pnl
PASS 37: NXPI PnL dates follow its own available price history
PASS 38: Vectorized median-of-three reference agrees with transparent NXPI loop
PASS 39: NXPI buy/hold and strategy performance ratios match independently
PASS 40: NXPI PnL prefix remains invariant when later data are absent
PASS 41: Future stock-price perturbation preserves installed trailing medians
PASS 42: Future stock-price perturbation preserves installed trailing MAD

Recomputing every eligible stock from independent median-of-three arithmetic...
PASS 43: All-stock aggregation aligns by calendar date
PASS 44: Available-stock counts match independently aligned histories
PASS 45: Every all-stock mean PnL matches independent date-aligned arithmetic
PASS 46: Full-sample stock Sharpe ranking matches independent universe calculation
PASS 47: Every individual-stock Sharpe matches independent PnLs
PASS 48: 2016/2017 sample split is disjoint and exhaustive
PASS 49: IS/OOS cohorts match the same finite-history stock symbols in the same order
PASS 50: All matched in-sample Sharpes recompute independently
PASS 51: All matched out-of-sample Sharpes recompute independently
PASS 52: Saved selection ranking uses all valid in-sample stock histories
```

```
PASS 53: In-sample selection ranking values are correct
PASS 54: Top 100 are selected only from independently recomputed IS Sharpes
PASS 55: Top-100 portfolio retains the correctly aligned calendar
PASS 56: Every top-100 portfolio PnL matches independent aligned arithmetic
PASS 57: Transparent median/mad loop validates full history of NXPI
PASS 58: Transparent median/mad loop validates full history of TMO
PASS 59: Transparent median/mad loop validates full history of LW
PASS 60: Transparent median/mad loop validates full history of MRNA
PASS 61: Reported IS/OOS correlation matches independent stock returns
PASS 62: Scatter regression coefficients match independent IS/OOS calculations
PASS 63: All portfolio-period sample sizes, Sharpes, cumulative PnLs and volatilities recompute

SUCCESS: 63 independent checks passed.
Verified 8352 SPY dates, 5042 stock dates, 717 eligible stocks and 645 matched IS/OOS stocks.
EMA causality checks exclude the package's two-observation startup; full-sample scaling is ex-post
    normalization.
```